

$$Q \pm \Delta Q = (1.1 \pm 0.1) \times 10^{-4} \text{ C}$$

- 2** A person supports a load of 20 N in his hand as shown in Fig. 2.1. The system consisting of the arm and load is equivalent to the system shown in Fig. 2.2. The rod represents the forearm, and T represents the tension exerted in the biceps. The forearm weighs 65 N.

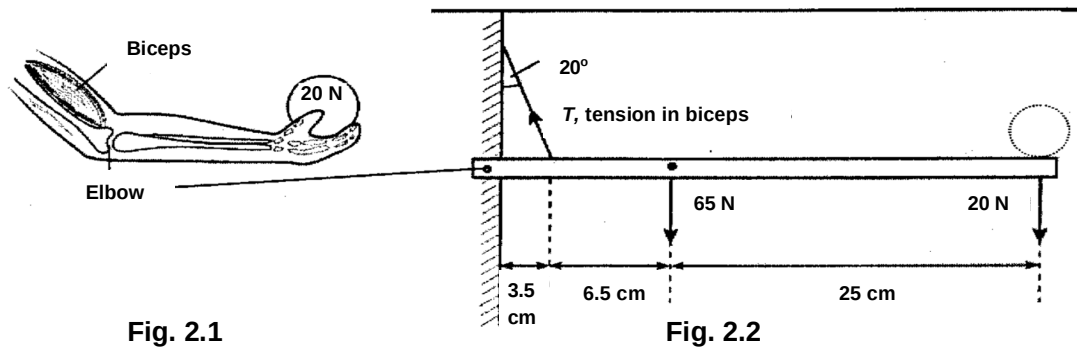


Fig. 2.1

Fig. 2.2

- (i) Show that the tension T in the biceps is 410 N.

[2]

Taking moments about elbow

$$T \cos 20^\circ \times 0.035 = [20 \times (0.250 + 0.065 + 0.035)] + [65 \times (0.065 + 0.035)]$$

$$T = 410 \text{ N}$$

- (ii) Hence, determine the magnitude and direction of the force acting at the elbow.

Let the force acting on the elbow be F .

Horizontal component of $F = 410 \times \sin 20^\circ = 140.2 \text{ N}$ right

Vertical component of $F = (410 \times \cos 20^\circ) - 65 - 20 = 300.3 \text{ N}$ downwards

$$F = \sqrt{(300.3^2 + 140.2^2)} = 331 \text{ N}$$

$$\theta = \tan^{-1}(300.3/140.2) = 65^\circ \text{ below forearm}$$

